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Long-term effects of forty-hertz auditory stimulation as a treatment of Alzheimer's disease: Insights from an aged monkey model study

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Based mainly on rodents studies, forty-hertz (40-Hz) physical stimulation has been regarded as a potential noninvasive treatment for Alzheimer's disease (AD). Considering the brain differences between rodents and humans, the effects of 40-Hz physical stimulation need to be further validated using nonhuman primates before its clinical application. Here, we took advantage of a rare opportunity to expose nine aged rhesus monkeys (26 to 31 y old) to 40-Hz auditory stimulation. Given the strong correlation between cerebrospinal fluid (CSF) Aβ and Tau concentrations and corresponding AD pathology in brain parenchyma in clinical practice, we investigated the effects of 40-Hz stimulation on AD pathology by monitoring changes in CSF Aβ and Tau concentrations. Our results revealed that 7 consecutive days of 40-Hz auditory stimulation triggered a rapid and significant increase of Aβ levels by more than 200%, but no effect on Tau levels in the CSF. Additionally, we observed that the elevation of CSF Aβ levels persisted for more than 5 wk after cessation, which had not been reported in any previous studies. After this, a pathological examination of the temporal cortices of 4 of the experimental monkeys was carried out and the data demonstrated that all of them had prevalent extracellular Aβ senile plaque pathology, whereas Tau pathology was negative or very weak. These results provide a good explanation for the differences between the CSF Aβ and Tau protein levels. Together, these first-time results from monkeys suggest that 40-Hz auditory stimulation has strong potential of a noninvasive AD treatment method.

Forty-hertz auditory stimulation | aged monkeys | Aβ deposition | Aβ levels | Alzheimer's disease

Significance

Although 40-Hz physical stimulation shows therapeutic potential for Alzheimer's disease (AD) in rodents, translational validation in nonhuman primates is critical. We applied 40-Hz auditory stimulation on nine aged rhesus monkeys and monitored their cerebrospinal fluid (CSF) Aβ and Tau level changes as indices of the treatment effect. Seven days' stimulation triggered a rapid CSF Aβ increase by more than 200%. In addition, the Aβ elevation persisted for over five weeks after treatment cessation-unreported in any rodent studies. Postmortem analysis of four of the experimental monkeys revealed widespread Aβ plaques in the brains. This study provides the first primate evidence of 40-Hz auditory stimulation can sustainably modulate the Aβ metabolism in the brain, supporting its potential as a noninvasive AD treatment method.

Alzheimer's disease (AD) is the world's leading neurodegenerative disorder and ranks number one among all types of dementia (1–3). Its symptoms are characterized by progressive loss of cognitive functions, particularly in learning and memory (4–6), and its pathology is characterized by the formation of amyloid senile plaques (SPs) and neurofibrillary tangles (NFTs) in the brain parenchyma (7–9). Despite extensive researches conducted worldwide, there was no effective treatment to reverse or even stop the progression of the disease. In recent years, three monoclonal antibody drugs (aducanumab, lecanemab, and donanemab) targeting Aβ have entered clinical trials and been approved by the US Food and Drug Administration (FDA). All these three drugs show only modest efficacy in early-stage AD patients and cannot reverse disease progression. After 18 mo of treatment, clinical symptom deterioration was slowed by approximately 30% compared to placebo (as measured by the Clinical Dementia Rating-Sum-of-Boxes, a commonly used clinical scale for assessing cognitive function and activities of daily living in AD patients) (10–12).With limited effects, they are all accompanied by high rates of side effects such as cerebral hemorrhage (6.8 to 22.1%) and cerebral edema (9.9 to 36%) (13). So, more effective and less side effect treatment is still urgently needed.

Forty-hertz (40-Hz) belongs to high-frequency electroencephalographic gamma oscillations generated by synchronized neural network activity in the brain, and they appear to be attenuated in the brains of AD patients and transgenic AD mouse models (14, 15). In 2016, a study by Tsai's group in MIT showed that 40-Hz light stimulation induced gamma oscillations, resulting in a decrease in amyloid levels and plaque loads in the primary visual cortex of transgenic AD mice, suggesting potential for noninvasive physical stimulation to treat AD (16). Later, the group extended the 40-Hz stimulation from visual to auditory by adding 40-Hz auditory stimulation to stimulate the temporal cortex (TC) and hippocampus, two key brain regions for learning and memory (17). The results demonstrated that the combined 40-Hz stimulation not only reduced plaque loads but also enhanced cognitive function, further highlighting the potential of 40-Hz for the

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treatment of AD. Similar studies from other groups have subsequently confirmed the therapeutic effects of 40-Hz stimulation (18–21). A recent mechanistic study by Tsai's team suggests that 40-Hz stimulation can enhance arterial pulsation and dilates meningeal lymphatic vessels to accelerate the influx of CSF into brain parenchyma and the efflux of interstitial fluid into CSF in AD mouse model, thereby removing Aβs from the brain through the glymphatic system (22).

While these findings support the idea of 40-Hz physical stimulation as a potential noninvasive treatment for AD, its feasibility for clinical translation still needs to be tested in more valid animal models. For example, drug candidates developed based on transgenic AD mouse models have failed to translate successfully in clinical trials in majority of the cases, casting doubts on the validity of AD rodent models in clinic translation (23–26). So far, all studies have been carried out on rodents, whose brain structures and functions are very different from humans. More importantly, mice do not naturally develop AD (27, 28), and only develop partial AD phenotype through artificial manipulation of familial AD genes that account for only 5% of all AD cases. In addition, a new study by Soula et al. reported contradictory results to those previous investigations. Using 40-Hz visual stimulation, they were unable to introduce gamma oscillations and did not detect a decrease of Aβ load in AD mouse brain (29). In other words, the latest result challenges the basis of 40-Hz therapy for AD.

In order to better investigate the therapeutic effects of 40-Hz physical stimulation for AD, we have resorted to nonhuman primates that have better relevance than rodents for the study of complex brain disorders like AD (30–35). The highly developed central nervous system is an important feature that distinguishes humans from other animals. Due to evolutionary proximity, rhesus monkeys and humans share more similar brain structures and higher functions than rodents. Many complex brain disorders thought to be unique to humans also occur naturally in monkeys. For example, our lab has reported the first spontaneous Parkinson's disease case in cynomolgus monkeys (32, 33), and the first naturally developed autism spectrum disorder rhesus monkey model (36). In term of monkey AD, since Wisniewski et al. published their pioneering study in 1973 (37), it has been widely accepted that SPs, one of the two most important AD pathological markers, spontaneously emerge in monkey brains with aging (38, 39). Regarding NFTs, the other most important AD markers, Hartig et al. first reported the phosphorylated Tau immunopositive neurons in aged mammals including rhesus monkeys and baboons (40), a finding later corroborated by Carlyle et al. (41), demonstrating age-dependent increases of phosphorylated Tau in macaque brains. However, conclusive evidence for NFT pathology in monkeys remained absent until 2017, when Arnsten's team provided rather clear evidence using immunoelectron microscopy combined with other techniques (42). Therefore, we believe that aged rhesus monkeys, which often naturally develop AD pathology, especially SPs, are ideal subjects to test the therapeutic effects of 40-Hz stimulation.

Compared with the occipital cortex (visual cortex), pathological changes of SPs and NFTs in AD patients are manifested much more strongly in the temporal cortex (auditory cortex), a brain region known for its susceptibility to AD. Therefore, in this study, we chose to stimulate aged rhesus monkeys with 40-Hz auditory stimulus, which would mainly activate the auditory pathway located in the temporal lobe. This is the first nonhuman primates study to investigate the efficacy of 40-Hz stimulation as a treatment for AD. Since aged monkeys are a precious and limited resource, a minimally invasive approach was adopted in this work by monitoring the changes of AD biomarker concentrations in cerebrospinal fluid (CSF) to illustrate the effects of the stimulation on AD pathology without killing the subjects for in clinical practice, the concentrations of Aβ and Tau in CSF are strongly correlated with Aβ plaque and Tau tangle pathology in brain parenchyma and has been used for AD diagnosis (43–46). Furthermore, previous studies have also demonstrated that 40-Hz auditory stimulation accelerates the influx of CSF into brain parenchyma and the efflux of interstitial fluid into CSF, thereby clearing AD-related pathological aggregates (22, 47). Therefore, we hypothesized that if 40-Hz stimulation could modulate Aβ and Tau metabolism in aged monkey brain tissues, we should be able to see a significant increase in their concentrations in CSF.

Results

Forty-Hertz Auditory Stimulation Rapidly Increased Aβ Levels in CSF. The results from the first round of stimulation for Group (a) showed that 7 d of continuous 40-Hz auditory stimulation rapidly increased Aβ levels in the CSF by twofold (205.61 ± 27.84% for Aβ42; 201.00 ± 20.98% for Aβ40), and further statistical analyses showed that these increases were significant (paired *t* test, two-tailed: Aβ42, $P = 0.0241$, Fig. 1*B*; Aβ40, $P = 0.0109$, Fig. 1*C*). In contrast, in the second round of random stimulation on Group (a) 21 d after the first round, Aβ levels in the CSF remained unchanged both before and after the stimulation (paired *t* test, two-tailed: Aβ42, $P = 0.3124$, Fig. 1*D*; Aβ40, $P = 0.2415$, Fig. 1*E*). Since the random stimulation was performed after the 40-Hz stimulation in Group (a), in order to control for possible sequential effects, the monkeys in Group (b) received the random auditory stimulation without proceeding 40-Hz stimulation. The results again confirmed that random stimulation did not affect Aβ levels in the CSF in Group (b) (paired *t* test, two-tailed: Aβ42, $P = 0.7681$, Fig. 1*F*; Aβ40, $P = 0.3403$, Fig. 1*G*). To exclude the possibility of the experimental procedure and animal handling in Group (a) (e.g. CSF collection and speaker setup) that might affect CSF AD biomarkers levels, in Group (c) was a control group that the speakers was placed in front of the monkeys' cages without playing any sound stimuli. The results showed that, in the absence of auditory stimulation, there were no significant changes in Aβ levels either in the 14 d of the first round (paired *t* test, two-tailed: Aβ42, $P = 0.2087$, Fig. 1*H*; Aβ40, $P = 0.2948$, Fig. 1*I*), or in the second round of test (paired *t* test, two-tailed: Aβ42, $P = 0.9180$, Fig. 1*J*; Aβ40, $P = 0.5770$, Fig. 1*K*).

To further demonstrate the effects of the two stimuli on Aβ levels, ΔAβs, the change between pre- and poststimulation (poststimulation Aβs concentration minus prestimulation Aβs concentration) were compared between the 40-Hz and random stimulation conditions of Group (a). Unsurprisingly, the ΔAβs were significantly higher in the 40-Hz stimulation than that in the random stimulation (ΔAβ42, $P = 0.0352$, *SI Appendix*, Fig. S1*A*, unpaired *t* test, two-tailed; ΔAβ40, $P = 0.0121$, *SI Appendix*, Fig. S1*B*, unpaired *t* test, two-tailed).

Detailed information on Aβs and ΔAβs levels before and after manipulation of three groups is presented in *SI Appendix*, Table S2-S5.

Together, these findings revealed that 40-Hz auditory stimulation rapidly induced significant 40-Hz specific increases in Aβ42 and Aβ40 levels by more than 200% in the CSF of aged rhesus monkeys.

Forty-Hertz Auditory Stimulation Had Long Term Effects On CSF Aβ Levels, Which Were Not Reported in Previous Studies. Our data also revealed another unexpected important result. Unlike studies based on rodents, where the effects of 40-Hz auditory

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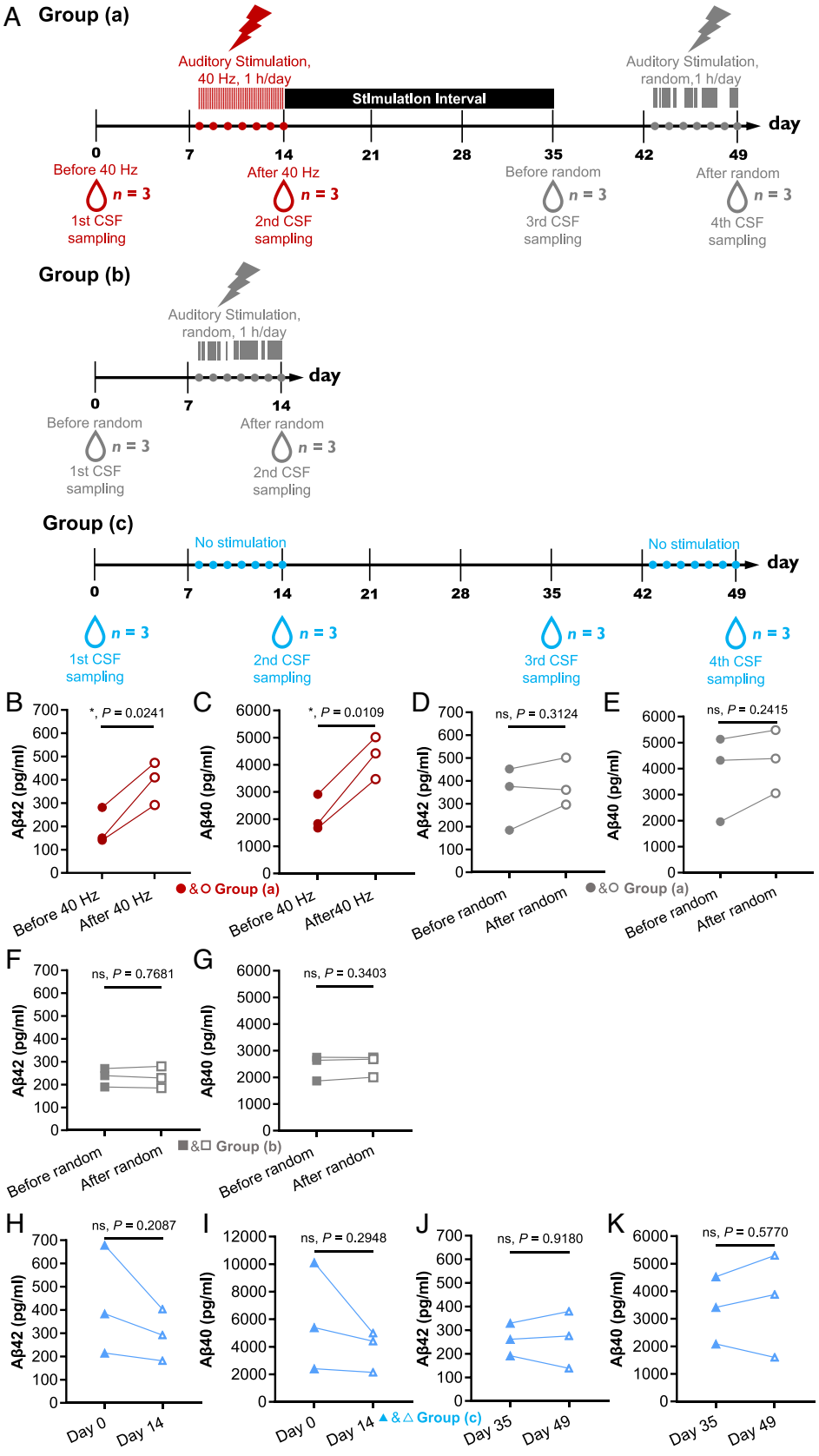

Fig. 1. Experimental design and the effects of 40-Hz [Group (a)], random [Group (b)] or no auditory stimulation [Group (c)] on CSF Aβ42 and Aβ40 concentrations in aged monkeys. (*A*) Experimental design and time points for CSF collection in Group (a), Group (b), and Group (c). Three aged monkeys in Group (a) were first subjected to 40-Hz auditory stimulation and then to random auditory stimulation, while three aged monkeys in Group (b) were subjected to random auditory stimulation only. To control for possible effects of handling animals overall in CSF biomarkers over the 49 d' experimental period, the three aged monkeys in Group (c) were subjected to the same procedures as those in Group (a), except that they were not exposed to any auditory stimuli (speakers were placed in the same location but no sound stimuli were played). All stimuli were administered continuously for 7 d (1 h/d). Water-drop symbols indicate time points of CSF sampling. The red color represents the 40-Hz manipulation round, including pre- and poststimulation CSF sampling and a 7-d period of auditory stimulation. The gray color represents random stimulation round and the blue color represents the no auditory stimulation round in Group (c) that there is a speaker but no sound stimulation. *n* indicates number of monkeys sampled for each CSF collection. Time interval between first and second CSF collection and between third and fourth CSF collection was 14 d. (*B* and *C*) CSF concentrations of Aβ42 and Aβ40 significantly increased (doubled) after 40-Hz auditory stimulation compared to before 40-Hz auditory stimulation in Group (a) ($n = 3$, $^*P < 0.05$). (*D* and *E*) CSF concentrations of Aβ42 and Aβ40 showed no significant changes after random auditory stimulation in Group (a) ($n = 3$, not significant, ns). (*F* and *G*) CSF concentrations of Aβ42 and Aβ40 showed no significant changes after random auditory stimulation in Group (b) (not significant, ns). (*H*–*K*) In the absence of auditory stimulation in Group (c) study, CSF concentrations of Aβ42 and Aβ40 showed no significant changes during the 14 d of both first and second rounds of manipulation ($n = 3$, not significant, ns).

stimulation disappeared 7 d after the stimulation, Aβ levels in aged rhesus monkeys remained at the increased levels for more than 35 d (5 wk). In Group (a), after 7 consecutive days of 40-Hz stimulation (day 14), CSF concentrations of Aβ42 and Aβ40 were more than doubled compared with the prestimulation baseline (day 0). Subsequently, the monkeys were in an intertrail interval for 21 d, and new CSF samples collected for a third time on day 35 after the baseline sampling, just prior to the random stimulation round. Interestingly, contrary to our expectations, the concentrations of Aβ42 and Aβ40 on day 35 were not significantly lower than those

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on day 14 (paired t test, two-tailed: Aβ42, P = 0.6103, Fig. 2*A*; Aβ40, P = 0.6687, Fig. 2*B*), indicating that the effect of 40-Hz auditory stimulation on monkeys did not disappear after 7 d, as previously reported on rodents (17), but still persisted. Since random stimulation did not change Aβ levels, we next compared samples from day 49 (after randomized stimulation) with those from day 14 (right after the 40 Hz stimulation) to further investigate whether this long-term effect persisted beyond day 35. The results showed that Aβ concentrations in CSF on day 49 did not change significantly from those on day 14. That is, the effect of 40-Hz stimulation on CSF Aβ42 and Aβ40 levels remained stable over the 35-day period (from day 14 to day 49, paired t test, two-tailed: Aβ42, P = 0.9324, Fig. 2*A*; Aβ40, P = 0.9953, Fig. 2*B*). Finally, we compared the CSF samples collected on day 49 with the baseline samples collected on day 0 (before 40-Hz stimulation) and the results showed that the concentrations of Aβ42 and Aβ40 were still more than doubled from the baseline, suggesting again that the effect of 40-Hz auditory stimulation in rhesus monkeys can last for more than 35 d (5 wk) (elevation of Aβ42: 202.82 ± 31.85%, elevation of Aβ40: 201.22 ± 32.78%, paired t test, two-tailed: Aβ42, P = 0.0155, Fig. 2*A*; Aβ40, P = 0.0451, Fig. 2*B*). To control for the animal handling and experimental procedure in Group (a) that might affect CSF biomarkers levels, in Group (c), a control group that was untreated with auditory stimulation for the same duration of time, CSF samples were collected on days 0, 14, 35, and 49, the same as in Group (a), and the results showed that the levels of Aβ in the CSF remained stable over the 49-d period (ANOVA: Aβ42, P = 0.5096, Fig. 2*C*; Aβ40, P = 0.5361, Fig. 2*D*).

These findings demonstrated that the effects of 40-Hz auditory stimulation on CSF Aβ42 and Aβ40 concentrations in aged monkeys lasted for more than 5 wk (day 14 to day 49), which had not been reported in previous studies using rodents.

Forty-Hertz Auditory Stimulation Had No Significant Effect On CSF Tau Levels. The results showed that the 40-Hz stimulation in the first round caused a slight increase in Tau levels in the CSF of Group (a), but following statistical tests revealed that the increases in both t-Tau and p-Tau 181 levels were not significant (paired t test, two-tailed: t-Tau, P = 0.2516, Fig. 3*A*; p-Tau 181, P = 0.4155, Fig. 3*B*). In the second round of random stimulation in Group (a), Tau levels in the CSF also remained unchanged after the randomized stimulation (paired t test, two-tailed: t-Tau, P = 0.9465, Fig. 3*C*; p-Tau 181, P = 0.7209, Fig. 3*D*). To control for possible order effects in Group (a), three additional aged monkeys in Group (b) received random stimuli at the beginning of the experiment. The results remained the same (paired t test, two-tailed: t-Tau, P = 0.7363, Fig. 3*E*; p-Tau 181, P = 0.9532, Fig. 3*F*). To see whether the handling and procedure in the Group (a) experimental duration could affect CSF biomarkers levels, Group (c) was exposed to the same procedure as in Group (a), except no auditory stimulation was played. Under this condition, there were no significant changes in Tau levels either over the 14 d before and after the first round test (paired t test, two-tailed: t-Tau,

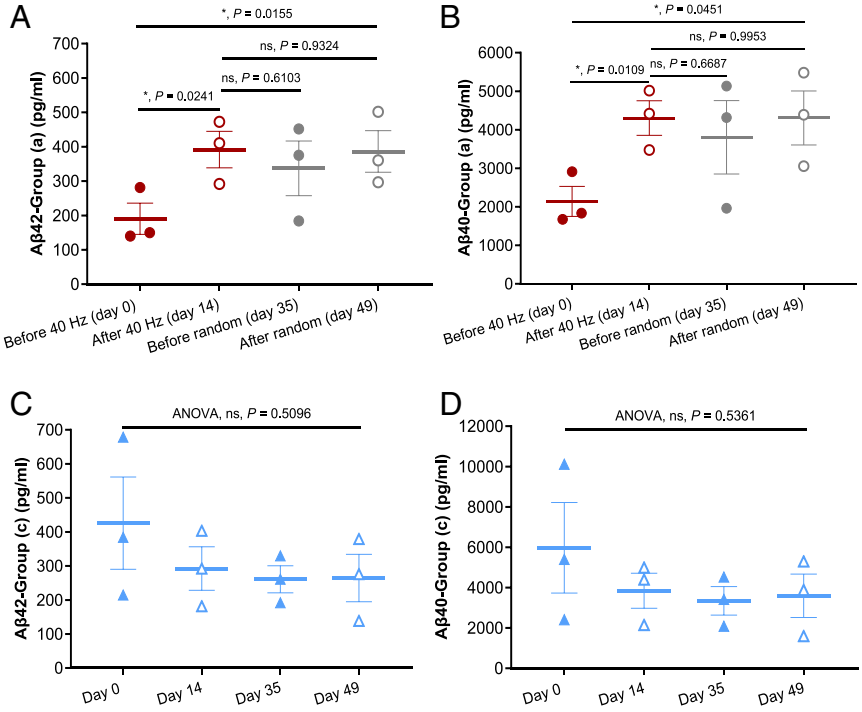

Fig. 2. Long-term effects of 40-Hz stimulation on CSF Aβ42 and Aβ40 levels of aged monkeys. In Group (a), the effects of 40-Hz auditory stimulation on CSF concentrations of Aβ42 (*A*) and Aβ40 (*B*) were long-lasting. Compared to the levels before 40-Hz auditory stimulation (day 0), CSF concentrations of Aβ42 and Aβ40 in the three monkeys of Group (a) were significantly increased after 40-Hz stimulation (day 14) (n = 3, *P < 0.05). After a 21-d stimulation interval, the CSF concentrations of Aβ42 and Aβ40 were not significantly changed on day 35 compared to day 14 (immediately after the 40-Hz stimulation) (n = 3, P > 0.05; not significant, ns), and remained stable at a high level. When data from the last CSF collection in Group (a) (day 49, the day right after the random stimulation) were compared to baseline (day 0), the CSF concentrations of Aβ42 and Aβ40 still showed a significant two-fold increase (Aβ42, 202.82 ± 31.85%; Aβ40, 201.22 ± 32.78%; n = 3, *P < 0.05). Comparison of the samples of day 49 and day 14 showed that CSF Aβ42 and Aβ40 levels remained unchanged after the 40-Hz stimulation (n = 3, P > 0.05; not significant, ns). In the absence of any auditory stimulation in Group (c) study, there were no significant changes in CSF Aβ42 (*C*) and Aβ40 (*D*) over a period of 49 d (n = 3, P > 0.05; not significant, ns). These results suggest that the effects of 40-Hz auditory stimulation on CSF concentrations of Aβ42 and Aβ40 lasted for more than 35 d. Data are mean ± SEM.

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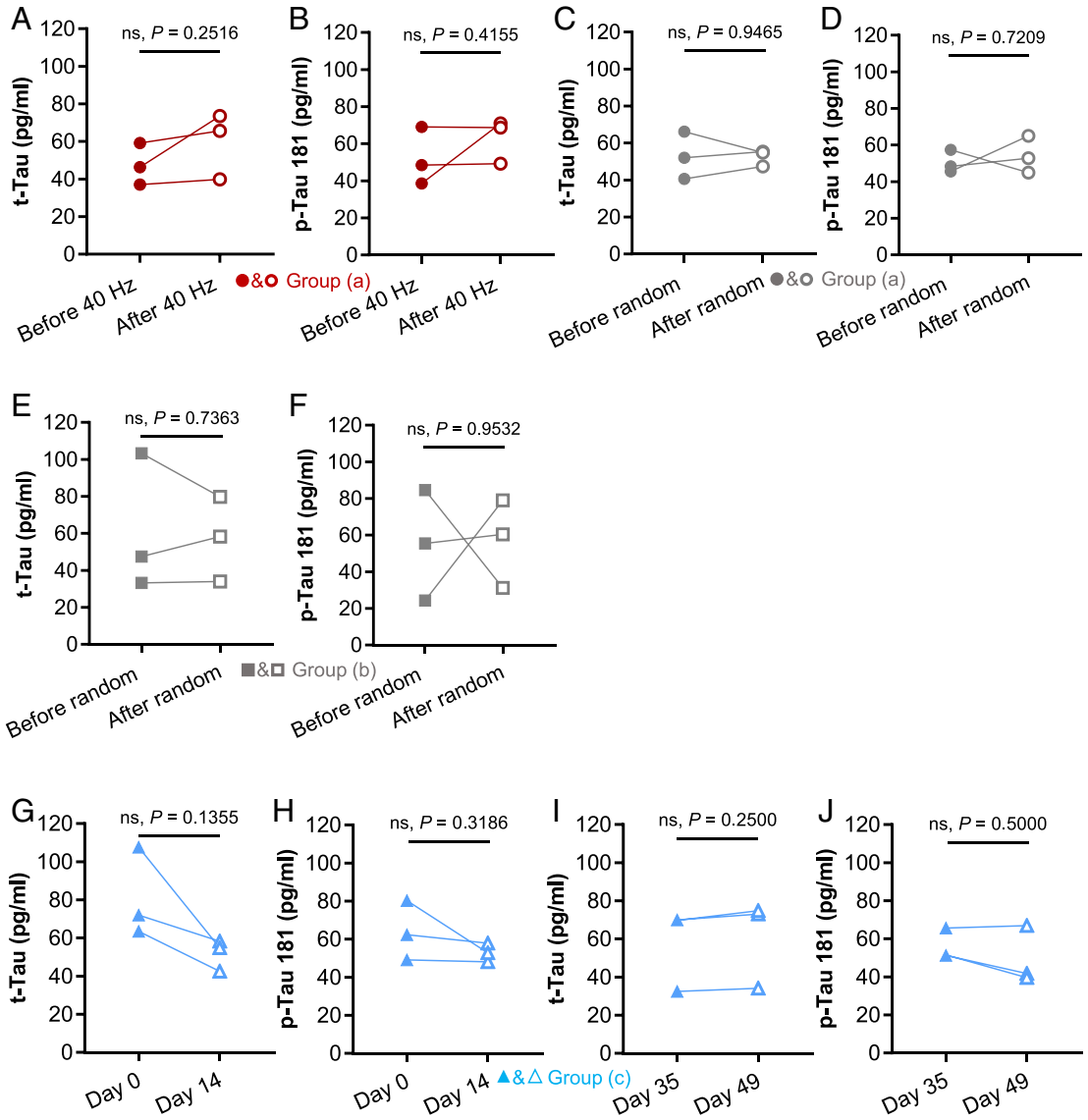

Fig. 3. The effects of 40-Hz [Group (a)], random [Group (b)] or no [Group (c)] auditory stimulation on CSF concentrations of t-Tau and p-Tau 181 in aged monkeys. (*A* and *B*) CSF concentrations of t-Tau and p-Tau 181 had a slight but not significant increase after 40-Hz stimulation compared to before the stimulation in Group (a) ($n = 3$, not significant, ns). (*C* and *D*) CSF concentrations of t-Tau and p-Tau 181 showed no changes after random auditory stimulation compared to before the stimulation in Group (a) ($n = 3$, not significant, ns). (*E* and *F*) CSF concentrations of t-Tau and p-Tau 181 showed no changes after random auditory stimulation compared to before random auditory stimulation in Group (b) ($n = 3$, not significant, ns). (*G–J*) In the absence of any auditory stimulation in Group (c) study, the CSF concentrations of t-Tau and p-Tau 181 did not change significantly over the 14 d of both first and second rounds of manipulation ($n = 3$, not significant, ns).

$P = 0.1355$, Fig. 3*G*; p-Tau 181, $P = 0.3186$, Fig. 3*H*), or the second round of test (paired *t* test, two-tailed: t-Tau, $P = 0.2500$, Fig. 3*I*; p-Tau 181, $P = 0.5000$, Fig. 3*J*). In addition, we also directly compared ΔTau (poststimulation Tau concentration minus prestimulation Tau concentration) for both the 40-Hz and randomized conditions of Group (a). The results showed that ΔTau was not significantly higher for the 40-Hz stimulation compared to random stimulation (unpaired *t* test, two-tailed: Δt-Tau, $P = 0.2531$, *SI Appendix*, Fig. S2*A*; Δp-Tau 181, $P = 0.6392$, *SI Appendix*, Fig. S2*B*). The finding that 40-Hz stimulation had no effect on Tau protein levels in CSF was consist with the negative Tau pathology in the brains of aged monkeys, which is presented in the next section.

Detailed information on Tau and ΔTau levels before and after manipulation of three groups is presented in *SI Appendix*, Table S2-S4 and S6.

These findings revealed that neither the 40-Hz auditory stimulation, nor random stimulation caused significant changes in CSF t-Tau and p-Tau 181 levels.

Aged Monkeys Have Obvious SPs But Negative/Very Mild Phosphorylated Tau Accumulation in Their Brains. Several studies have demonstrated that typical SPs and occasionally NFTs occur in the brain of most aged rhesus monkeys (37–42). Changes in CSF biomarkers are highly correlated with the development of AD pathology. Therefore, there should be a pathological basis for the above observed different effects of 40-Hz stimulation on Aβ and Tau levels in the CSF of aged rhesus monkeys. To test this hypothesis, at the end of the experiment, four of the six monkeys [monkeys 1, 2, 4, and 5, two from Group (a) and two from Group (b)] were euthanized [in accordance with the Guide for the Care and Use of Laboratory Animals (8th edition, 2011; National Academies Press, Washington, DC)]. Monkey 6 from Group (b) died suddenly at the end of the study due to uterine hemorrhage, while Monkey 3 from Group (a) participated in another experiment.

6E10 immunohistochemical staining identifies Aβ peptides and plaques. The staining revealed SPs of varying degrees of severity in

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the temporal cortex (TC) of all four monkeys. The TC of Monkeys 1 and 4 contained severe SPs, including both dense plaques (red arrows) and diffuse plaques (black arrows) (Fig. 4 *A* and *C*). Monkey 2 and Monkey 5 had diffuse plaques (Fig. 4 *B* and *D*). However, Tau pathology illustrated by AT180 immunohistochemical staining, which identifies aggregates of phosphorylated Tau, revealed that the neurons in the TC of the four aged monkeys had either lacked (Monkeys 1, 2, and 4, Fig. 5 *A*–*C*) or very slight positive staining (Monkey 5, Fig. 5*D* and *E*), suggesting either no or very low levels of phosphorylated Tau in the brains of these monkeys.

Discussion

In this study, we conducted the first investigation into the therapeutic effects of 40-Hz auditory stimulation using macaque monkeys, yielding two conclusions. First, our results showed that 40-Hz auditory stimulation could induce a rapid (within 7 d) and 40-Hz specific significant elevation of Aβ concentrations in the CSF of aged rhesus monkeys, indicating that the stimulation could affect Aβ metabolism by promoting extracellular Aβ efflux into CSF. Second, and the most important and unexpected finding in this study, the elevation of Aβ protein concentrations in CSF induced by the 40-Hz auditory stimulation lasted more than 5 wk while the possibility of animal handling and experimental procedure on the CSF Aβ level had been excluded by a control group that was untreated for the same duration of time. This long-term effect has not been reported in any of the previous studies using rodents, indicating there are significant Aβ metabolism differences between rodents and monkeys, though the underline mechanism needs to be further investigated.

The two most important pathological features in AD patients are the extracellular deposition of Aβ proteins which form SPs and the intracellular accumulation of phosphorylated Tau proteins which form NFTs. The pathological changes and metabolic abnormalities of these two proteins in the brain parenchyma are reflected in their CSF levels (48, 49). In recent years, abnormal concentrations of Aβ and Tau proteins in CSF have been widely used as biological markers for AD diagnosis (43–46). How does 40-Hz auditory stimulation cause an increase in CSF Aβ levels in the aged monkeys in this study? Our results is consistent with a new mechanism reported recently by both Tsai's team (22) and Sun et al. (47). The studies reveal that 40-Hz physical stimulation can cause an increase in astrocyte aquaporin-4 polarization, enhancement of arterial pulsatility, and dilation of meningeal lymphatic vessels, thereby promoting the CSF flow into the brain parenchyma and clearance of metabolic waste, including Aβ, to the CSF. Although we hypothesize above based on literatures (22, 47) that elevated Aβ levels in CSF result from enhanced glymphatic clearance following 40-Hz stimulation, we have not directly measured glymphatic function or interstitial Aβ efflux. In future studies, we can directly observe 40-Hz induced efflux through two approaches. First, by injecting isotope-labeled Aβ into monkey brain parenchyma, we can measure labeled Aβ concentrations in CSF after 40-Hz stimulation, directly observing Aβ clearance rates. Second, using implanted lumbar puncture needles, we can continuously collect and measure CSF Aβ levels before, during, and after 40-Hz stimulation, directly observing the Aβ efflux effect induced by 40-Hz stimulation.

Why no tau concentration changes was observed in CSF of this study? In the subsequent pathological tests, we found that 4 out of the 6 aged monkeys treated with auditory stimulation had significant SPs pathology in the temporal region in which the auditory cortex is located, while phosphorylated tau deposition pathology was either absent or only very mildly presented. We speculate that 40-Hz auditory stimulation promotes the flush of Aβ proteins from

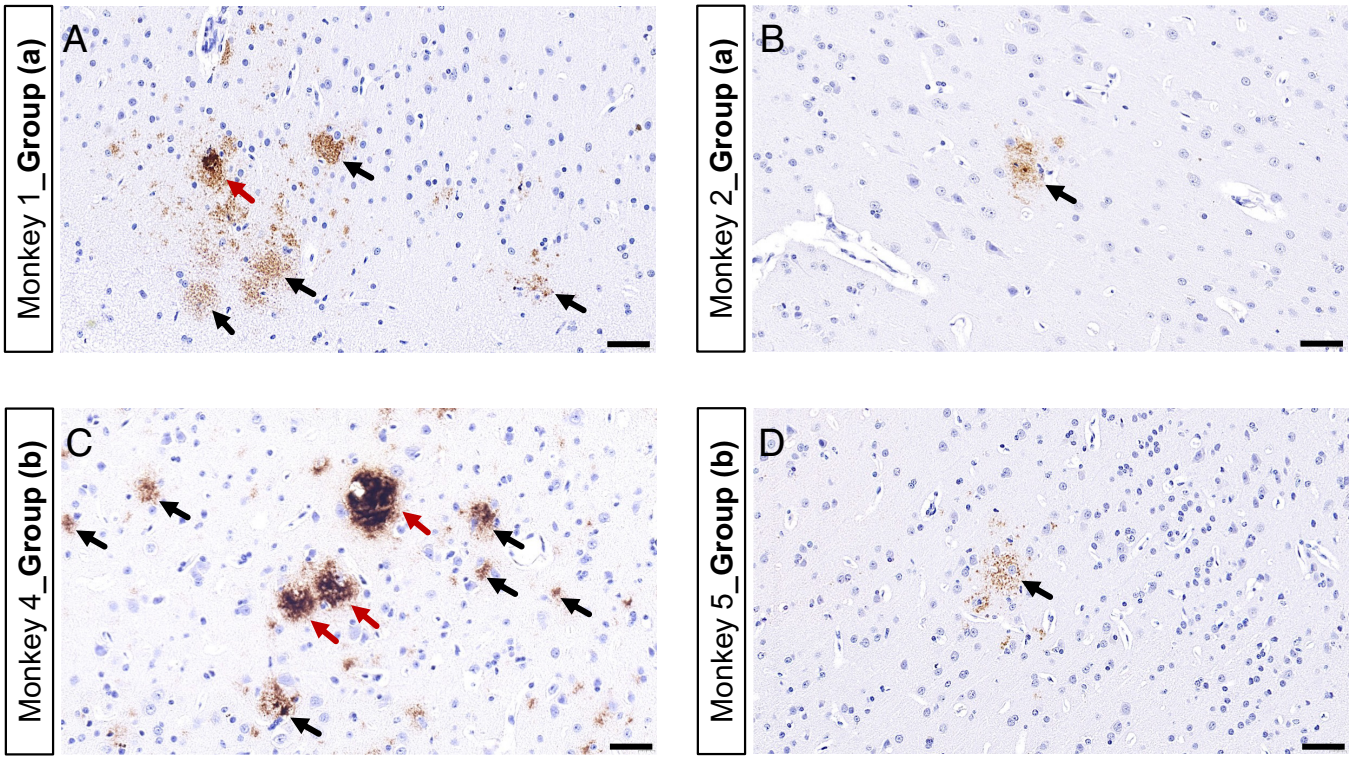

Fig. 4. Representative images of Aβ pathology stained by 6E10 in TC of four aged monkeys. TC Aβ plaque loads in four aged monkeys [two of each of Group (a) and Group (b)]. (*A* and *B*) Monkey 1 and 4 had a great number of Aβ plaques, including both dense (red arrows) and diffuse (black arrows) plaques. (*C* and *D*) Monkey 2 and 5 had a few diffuse plaques (black arrows). (Scale bar, 50 μm.)

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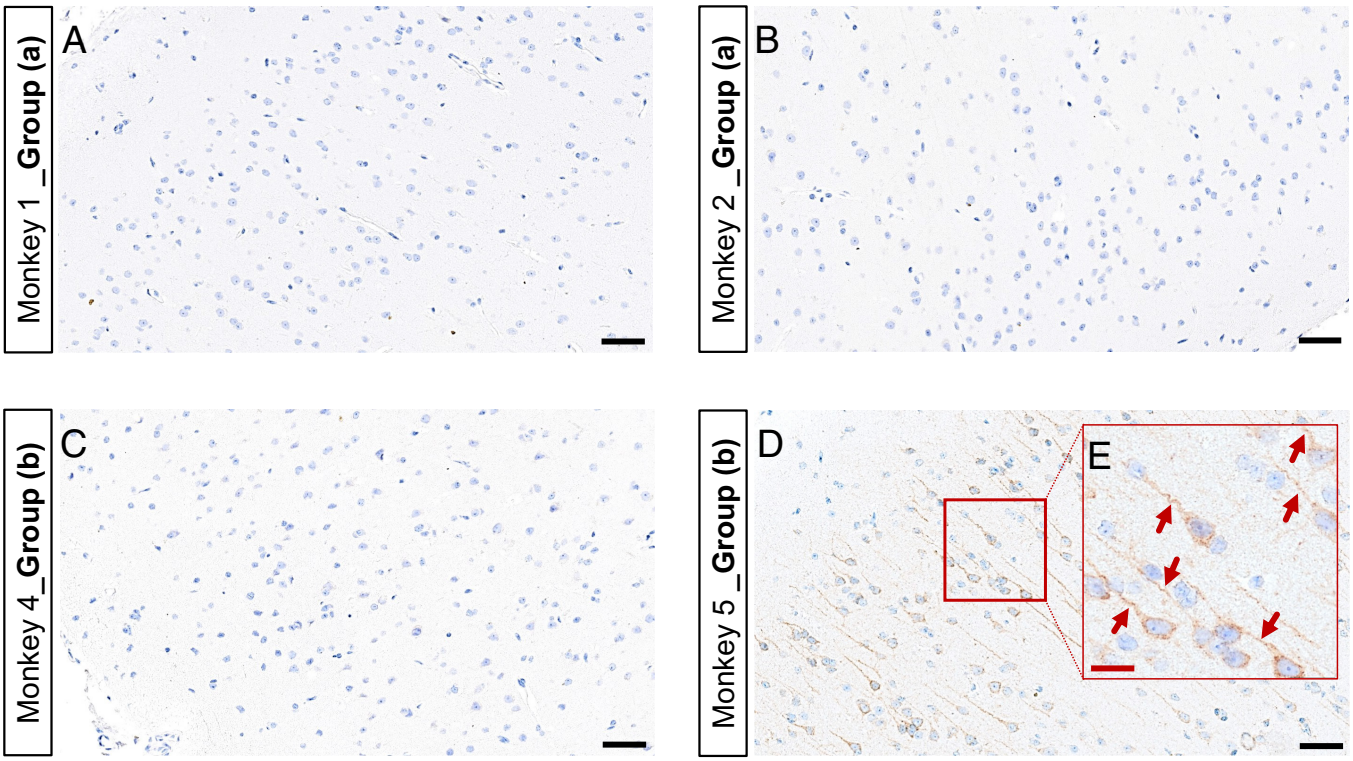

Fig. 5. Representative images of Tau pathology stained by AT180 in TC of four aged monkeys. TC phosphorylated Tau accumulation in four aged monkeys [two of each of Group (a) and Group (b)]. (*A*–*C*) Monkeys 1, 2, and 4 had no positive signals indicating there was no phosphorylated Tau accumulation. (*D* and *E*) Monkey 5 had very mild phosphorylated Tau deposition in the soma and neurites (red arrows). (Scale bar, 50 μm in *A*–*D* and 20 μm in *E*.)

the intercellular fluid to the CSF through activation of the glymphatic system, which in turn leads to the elevation of the CSF Aβ protein levels. On the other hand, since there was no significant accumulation of phosphorylated Tau in the brains of the aged monkeys, no Tau level changes were observed in the CSF.

In conclusion, this first monkey study demonstrates the long-lasting positive effect of 40-Hz auditory stimulation on aged brain Aβ metabolism, and suggests that 40-Hz auditory stimulation has potential as a new non invasive AD treatment method.

Materials and Methods

Animals and Ethics. Nine aged rhesus monkeys (four males and five females, 26 to 31 y old) were purchased from the Experimental Animal Center of the Kunming Institute of Zoology (see *SI Appendix*, Table S1 for details). During the experiment, the rhesus monkeys were single-caged and exposed to 12-h light/dark cycles (07:00 am to 19:00 pm) under 60% humidity and at 21 ± 2 °C room temperature conditions. They were fed twice with monkey chow and once with fruit daily, with ad libitum access to water. Throughout the experiment, experienced veterinarians carefully monitored the health of these aged rhesus monkeys.

All experimental protocols had been approved by the Ethics Committee of the Kunming Institute of Zoology, Chinese Academy of Sciences (IACUC19017). All animals were treated and cared for in accordance with the Guide for the Care and Use of Laboratory Animals (8th edition, 2011; National Academies Press, Washington, DC) and updated ARRIVE guidelines (PLoS Biol. 2020;18(7): e3000411).

Experimental Design. According to our and other's previous studies, aged rhesus monkeys naturally develop amyloid plaques and sometimes phosphorylated Tau and they accumulate mainly in the temporal rather than in the visual lobe. In order to effectively investigate the effect of 40-Hz physical stimulation on AD pathology, we chose to perform auditory stimulation on aged rhesus monkeys, which would activate the temporal cortex Nine aged rhesus monkeys were randomly assigned to three groups (Fig. 1*A* and *SI Appendix*, Table S1). Group (a) consisted of three aged rhesus monkeys, two males and one female. The monkeys received one round of 40-Hz auditory stimulation followed by another round of random auditory stimulation with an interval of 21 d in between. Group (b) also consisted of three aged rhesus monkeys, one male and two females. The animals received only one round of random auditory stimulation to control for possible sequential effects in Group (a), in which the random stimuli were administered after the 40-Hz stimulation. Group (c) included three aged rhesus monkeys as well, one male and two females, designed to control for the animal handling and experimental procedure, which might affect CSF AD biomarkers levels, during the 49 d experimental time period of Group (a). Thus, the experimental manipulation of Group (c) was identical to all aspects of Group (a) except that there was no auditory stimulation (both random and 40-Hz) during the stimulation round (the speakers were set up but no auditory stimulus was played). In each round of stimulation, the rhesus monkeys were given 1 h of auditory stimulation at a fixed time (10:00 to 11:00 a.m.) in every day for 7 consecutive days. CSF samples were collected from the monkeys before and after each round of stimulation (Fig. 1*A*). By examining changes of Aβ and Tau levels in the CSF samples, the effects of the two types of auditory stimulation (40-Hz and random) on AD related pathological biomarkers were investigated. After the auditory stimulation, four out of the nine aged monkeys were euthanized [in accordance with the Guide for the Care and Use of Laboratory Animals (8th edition, 2011; National Academies Press, Washington, DC)] and pathological Aβ and Tau hallmarks in their brain were illustrated.

Auditory Stimulation. Based on the auditory stimuli used in previous studies using transgenic AD mouse model (17) and taking into account the optimal hearing range of rhesus monkeys (1 to 8 kHz) (50), we designed two types of auditory stimuli, 40-Hz and random stimuli. The 40-Hz auditory stimuli consisted of 1-kHz pure tones with a duration of 1 ms at a frequency of 40-Hz (i.e., one sound in every 25 ms) at an intensity of 60 dB. The random auditory stimuli consisted of 1-kHz pure tones of 1 ms duration at random intervals, and the average of the intervals was 25 ms and its intensity at 60 dB.

The rhesus monkeys were housed in a quiet room (4.2 × 5.7 × 6.2 m). Prior to the start of the experiment, the average background sound level was measured on three consecutive days at 10:00 a.m., 10:30 a.m., and 11:00 a.m. and

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calculated to be 36.5 ± 2.6 dB. This value showed that the background sound level was stable and quiet. After the start of experiment, the auditory stimulation was administrated every day from 10:00 a.m. to 11:00 a.m. for seven consecutive days as a single round. Two loudspeakers were placed in front of three square cages (side length 80 cm) at a distance of 42 cm from the cages. Inside the cages, rhesus monkeys received 40-Hz [Group (a)], random [Group (b)] or no stimuli [Group (c)] according to the experimental design (Fig. 1*A*).

CSF Sample Collection. CSF samples were collected before and after subjecting the aged rhesus monkeys to 40-Hz [Group (a)], random [Group (b)] or no [Group (c)] auditory stimulation for seven days (Fig. 1*A*). All CSF samples were collected between 14:00 and 15:00 p.m.

The CSF samples were collected via lumbar puncture by experienced experimenters. The CSF collection procedure was as follows (35). Following a general anesthesia induced by intramuscular injection of 4 mg/kg of Zoletil-50 (Virbac S.A., France), the shaved skin around the lumbar region was cleaned and disinfected with iodophor and 75% alcohol. The monkey was positioned naturally on the operating platform, with its back arched to expose the lumbar vertebrae. A sterilized stainless steel puncture needle (22 G, Shanghai Asclepius Medical Plastic Products Co., Ltd., China) was then inserted at a 90-degree angle to the lumbar vertebrae. The first drop of CSF was discarded. From each rhesus monkey, 1.2 mL of clear CSF was collected into a sterile polypropylene tube and rapidly mixed with a pipette. Then CSF samples were aliquoted into 0.5-ml sterile polypropylene cryogenic vials, quickly frozen in liquid nitrogen, and stored at −80 °C.

CSF AD Biomarker Analyses. Using highly sensitive single molecule array (Simoa) technology (Quanterix, MA), CSF Aβ42, Aβ40, total Tau (t-Tau), and phosphorylated Tau (p-Tau 181) concentrations before and after the experimental manipulation were assessed. Specifically, the Neurology 3-Plex A (Quanterix, MA, product number: 101995) and p-Tau 181 (Quanterix, MA, product number: 104618) kits were used on the automated Simoa HD-X platform (GBIO, Hangzhou, China). All samples were processed on the same 96-well plate using the same batch of assay kit. Each sample underwent duplicate testing, and the results were averaged.

AD Pathology Investigation. Following the auditory stimulation experiments, 4 monkeys (two from Group (a) and two from Group (b), Monkey 6 died suddenly at the end of the study due to uterine hemorrhage, while Monkey 3 participated in another experiment) were euthanized and immunohistochemical staining was performed with 6E10 (for the identification of Aβ peptides and staining of Aβ plaques) and AT180 (for the identification of phosphorylated Tau proteins and staining of cells with phosphorylated Tau accumulations) to demonstrate Aβ and Tau pathological conditions in the TC.

Specifically, the monkeys were first anesthetized with ketamine (10 mg/kg, intramuscular injection) and subsequently euthanatized with an overdose of pentobarbital sodium (100 mg/kg, intramuscular injection) and perfused with 2 L of 0.9% NaCl solution through the heart. Brain tissue was extracted and fixed in 4% paraformaldehyde (PFA) in 0.01 M phosphate-buffered saline (PBS) at 4 °C for one week. The fixed TC tissue samples were then coronally sectioned into 3-mm-thick blocks, embedded in paraffin, and cut into 4-μm-thick sections using a Leica RM2016 Microtome (Leica, Shanghai, China) for staining. The primary antibody used for immunohistochemical staining was 6E10 (1:200, EMD Millipore, Cat. #MAB5206) followed by goat anti-mouse IgG secondary antibody (1:200, Abcam, UK, Cat. #ab6789) and AT180 (1:200, Invitrogen, Cat. #MN1040) followed by Biotinylated Secondary Antibody (1:200, Vector Laboratories, Cat. #PK-4000).

For immunohistochemical staining, the paraffin sections were placed on a glass slide and baked overnight at 65 °C. The sections were deparaffinized and rehydrated twice (10 min each) with xylene and graded ethanol. The processed sections were then subjected to antigen retrieval in citrate buffer by heating in a pressure cooker for 2 min. After treating the sections with 70% methanol for 7 min, endogenous peroxidase was blocked by incubating the sections with 3% hydrogen peroxide in methanol for 10 min at room temperature. Subsequently, the sections were blocked in 2% bovine serum albumin (BSA, 0.5% Triton X-100) at 37 °C for 30 min. The primary antibody, diluted in blocking solution, was added, followed by overnight incubation at 4 °C. The sections were then treated with secondary antibody at room temperature for 2 h (for 6E10) or overnight at 4 °C (for AT180) and visualized using DAB kit (for 6E10, MXB, China, Cat. #DAB-1031) or ABC-HRP Kit (for AT180, Vector Laboratories, Cat. #PK-4000). Hematoxylin (Servicebio, China) was used to counterstain the nuclei. In the negative control experiment, no primary antibody was added. Finally, the sections were dehydrated and cleared with graded ethanol and xylene, mounted with a neutral resin, and covered with a coverslip. Pathological examination was performed using the glass-slide scanning imaging system SQS-40P (Shengqiang Technology, Shenzhen, China) for 6E10 stained sections, and VS200 (Olympus, Tokyo, Japan) for AT180 sections.

Statistical Analysis. For normally distributed data, paired *t* tests were used to analyze differences in CSF AD biomarker levels before and after the manipulation of Group (a), (b) and (c), as well as the long-term effects of 40-Hz auditory stimulation of Group (a). Unpaired *t* tests were used to analyze the changes of ΔAβ42 and ΔTau levels between two stimulation conditions of Group (a). ANOVA was used to analyze changes in CSF Aβ in Group (c) over 49 d. Only the t-Tau and p-Tau 181 levels of Day 35 and 49 of Group (c) did not show a normal distribution and Wilcoxon matched-pairs signed rank test was used (Fig. 3 *I* and *J*). For all analyses, $P < 0.05$ was considered significant (two-tailed). All statistical analyses were performed using GraphPad Prism 8.0 software (Dotmatics).

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*.

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The authors declare no competing interest.

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